

Technical University of Cluj-Napoca

PHD REPORTING

**Research Study on
End-to-End Autonomous
Driving**

**Research Theme: Environment Perception for
Intelligent Vehicles Motion Planning in Highly
Dynamic Scenarios**

Submitted by:

**Eng. Bianca-Veronica
AVRAM**

Supervisor:

**Prof. dr. eng. Florin
Ioan ONIGA**

December 2, 2025

Contents

1	Introduction	2
2	Perception	3
2.1	3D Object Detection	4
2.2	Map Generation	7
2.2.1	Occupancy Grids	8
2.2.2	Vectorized Scene Representation	10
3	Motion Forecasting and Trajectory Prediction	13
4	Planning-Oriented Autonomous Driving	17
5	Future Research Directions	21
5.1	Sensor Fusion	21
5.2	Network Optimization for Real-Time Capabilities	22
5.3	Challenging Scenarios	22
6	Final Remarks	22

1 Introduction

Autonomous driving has become a highly researched field, reflecting the high desire of car manufacturers to provide certain levels of autonomy for their customers. In this context, the objective is to develop a system that can assist or entirely replace the human driver while maintaining strict compliance with driving rules and ensuring passengers' safety.

Hence, the automotive industry strives to keep pace with the rapidly evolving technologies in environmental perception, artificial intelligence, and embedded systems. This study is a review of state-of-the-art methods for environment perception and path planning for autonomous vehicles.

Multiple approaches can be considered when designing such a system. Generally, a vehicle must be equipped with intelligent sensorial devices, such as cameras, radars, LiDARs, and ultrasounds. Then, there is an algorithm that interprets the measurements and decides the best and safest possible actions that the vehicle must perform, which are translated into control signals as steering, braking, and acceleration. The algorithm itself was designed in multiple versions in the industry, but there have been mainly two approaches in the literature that can be seen as two autonomous driving paradigms. On one hand, there is the modular pipeline (figure 1), a method that separates different stages from sensorial input to the control signals. This pipeline usually includes modules like perception, scene representation, motion prediction, and planning. On the other hand, there is the end-to-end pipeline (figure 2), which directly predicts either a trajectory for the ego vehicle or even directly the control signals necessary to drive the vehicle.

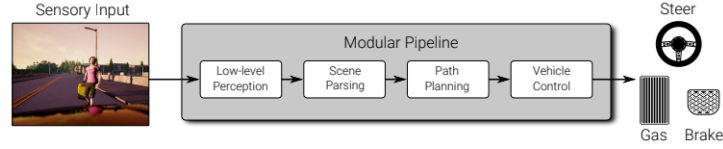


Figure 1: Modular Pipeline for Autonomous Driving [16].

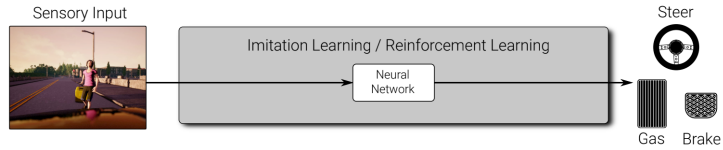


Figure 2: End-to-end Pipeline for Autonomous Driving [16].

Studies have shown that completely breaking the pipeline into submodules improves per-task performance, but it falls into the trap of overoptimizing single stages, disregarding the end goal of trajectory prediction. This translates into a lower generalization power and a higher risk of misdetections due to the

propagation of errors along the whole pipeline. At the same time, fully end-to-end approaches tend to be too dependent on training data due to the "black box" setup, which can also be a barrier towards validating such a model, due to insufficient and improper ground truth for the driving task.

As a consequence, literature started to explore hybrid systems and proposed a "planning-oriented" approach [19] (figure 3), a modularized version in which each module (perception, prediction, and planning) is trained in a task-specific manner, but benefits from training time interaction with the others. This technique avoids single-task optimization and ensures an abstract level of communication via modern query usage as intermediary representations for the scene elements.

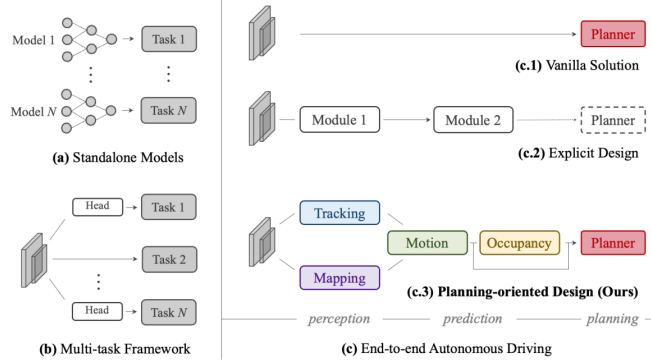


Figure 3: UniAD proposal for planning-oriented autonomous driving [19].

In this work, we will explore the components of a performant end-to-end autonomous driving system, emphasizing the state-of-the-art methods in the literature. We will cover the whole pipeline mentioned before, from perception towards trajectory prediction, but always having in mind the end goal of the proposed research theme. In the end, we will draw some directions for further research in the domain, based on the observed limitations in the current solutions.

2 Perception

In this section, we will address the problem of perception in autonomous driving. This step is a key towards scene understanding, for further learning of how to interact with the driving environment, which mainly consists of vehicles, vulnerable road users, obstacles, and infrastructure-related elements. All the methods presented in this chapter have as a main perception sensor a video camera or a set of cameras, and some additionally use point cloud data to perceive depth, either from LiDAR or from radar.

We will note papers that propose efficient solutions for 3D object detection and for map generation, two representative tasks for end-to-end autonomous driving.

From these papers, we emphasize the most remarkable contributions that will further facilitate the end-to-end goal of this research study.

2.1 3D Object Detection

Leveraging attention [46] in computer vision [13] has enabled object detection to take a step forward with increased accuracy and great potential for understanding complex scenarios, due to the high level of interaction between features proposed by the initial attention formula. DETR [6] introduced transformer architecture usage in object detection on images, and Deformable DETR [54] came with an enhanced learning mechanism that ensures faster convergence by using deformable attention, shown in equation 2. The speed enhancement is given by the fact that attention is not computed between every query and every key, but between each query, which becomes a reference point, and a limited number of keys, depending on their position relative to the reference point.

$$\text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (1)$$

where

$\text{Attn}(\cdot)$ is the aggregated features matrix.

Q, K, V are the **Query**, **Key**, and **Value** matrices, respectively.

$QK^\top$ is the dot-product score measuring compatibility between queries and keys.

$\sqrt{d_k}$ is the scaling factor (square root of the key dimension), used to stabilize gradients during softmax.

$\text{Softmax}(\cdot)$ converts raw scores into attention weights (a probability distribution).

$$\text{Attn}(Q_i) = \sum_{k=1}^K \alpha_{i,k} \cdot V(p_q + \Delta p_{qk}), \quad (2)$$

where

$\text{Attn}(Q_i)$ is the aggregated feature for the i -th object query.

K is the fixed number of sampling points (a small constant, e.g., 4) used for each query.

$\alpha_{i,k}$ is the attention weight for the k -th sampled point.

$V(\cdot)$ is the **value** feature sampled from the image feature map.

p_q is the reference point in the feature map (e.g., center of the query’s bounding box).

Δp_{qk} is the learned 2D offset for the k -th sampling point relative to the reference point.

To further deploy such detection methods in autonomous driving, 3D perception is crucial for decision-making. A straightforward solution would be to project the predicted objects from the image space to vehicle space. However, this introduces additional computations and mainly requires enhanced sensor capabilities for depth estimation.

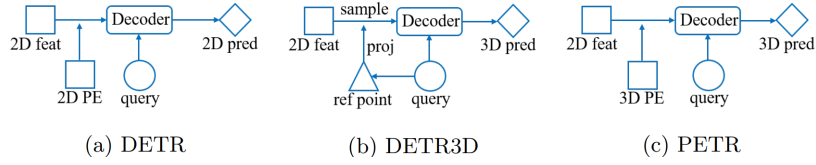


Figure 4: PETR comparison of 3D object detection [39].

To address this issue, DETR3D [47] proposes a 3D object detection framework using transformers. It introduces the use of object queries in a 3D space, which are then projected to image space via cameras’ parameters, and matched with features from 2D to refine the detections. To overcome projection errors, PETR [39] adds the 3D positional embeddings, enabling 3D object queries to avoid expensive perspective projection. A comparison of different detection paradigms in the mentioned articles is presented in figure 4. After this, PETRv2 [40] proposes an enhanced 3D positional embedding and adds temporal awareness as shown in Figure 5. They also employ a multi-task setup that additionally predicts map segmentation and lanes, but that will be explored in the next section.

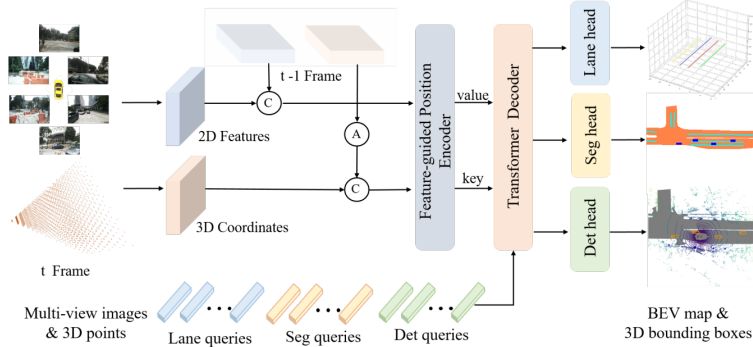


Figure 5: PETRv2 architecture [40].

Similarly, SparseBEV [37], from the same authors as PETR [39], comes with an enhanced attention mechanism, as in equation 3, that is actively adapted in accordance with the type of agent they represent, namely, it restricts query interaction to a certain receptive field, denoted by τ , depending on the nature of the instance interaction. This feature brings faster convergence time for the network and boosts performance at inference time. Still, it is conditioned by the dense BEV grid structure, which fails to model complex detections and is dependent on the dimensions of the grid in terms of computational efforts.

$$\text{Attn}(Q, K, V) = \text{Softmax} \left(\frac{QK^\top}{\sqrt{d}} - \tau D \right) V \quad (3)$$

Additionally, fully sparse approaches have been explored by the Sparse4D series [34, 35, 36]. Their main idea is to rely completely on a sparse detection by initializing a set of sparse queries that do not require any BEV grid structure. This brings even more speed while keeping up with the accuracy standards on public benchmarks (NDS [4]). SparseBEV uses pillar queries instead of points to incorporate better spatial priors, while Sparse4D’s queries use 4D keypoints as anchors inside the 3D detections to predict bounding boxes. Both include temporal information, but Sparse4D uses an advanced method based on recurrence, which enables better temporal consistency over a longer time period. Still, as a general issue of sparse query approaches, they have problems in catching fine-grained details in the scene. Furthermore, the initialization of these learned queries has a pronounced and crucial impact on the convergence speed and overall stability of the network during the training process.

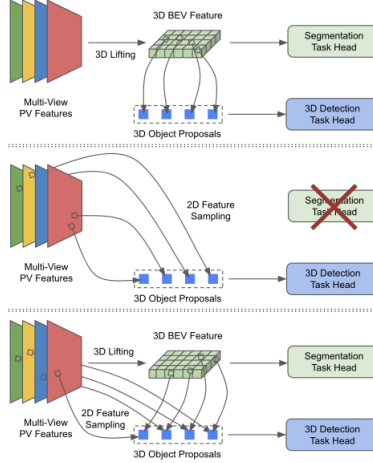


Figure 6: Comparison of different image-only 3D perception frameworks [23].

As seen in the previously analyzed works, there is a gap between dense and

sparse object detection. Recently, DuoSpaceNet [23] elaborates a valuable overview of 3D object detection methods that use dense grid-based queries and methods that use fully sparse queries, as seen in figure 6. This paper aims to establish a connection between BEV (bird’s-eye-view) based detection and PV (perspective-view) based detection, encompassing the map segmentation task simultaneously. It combines both BEV and PV embeddings in a single query, rather than separating them into two decoders. This empowers the newly created decoder to harmonize between the geometric features associated with the 2D to 3D transformations and the semantic features associated with the segmentation task. On top of that, they propose an elegant way of performing temporal training by introducing the temporal queries, capable of modelling both types of features. The authors also provide a comparison of metrics to the other state-of-the-art methods that have also been analyzed in this study in table 1.

Method	Image Size	Frames	mAP $\uparrow$	NDS $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	mAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
DETR3D [47]	1600 $\times$ 900	1	0.349	0.434	0.716	0.268	0.379	0.842	0.200
PETR [39]	1600 $\times$ 900	1	0.370	0.442	0.711	0.267	0.383	0.865	0.201
BEVFormer-S [30]	1600 $\times$ 900	1	0.375	0.448	0.725	0.272	0.391	0.802	0.200
Sparse4D [34]	1600 $\times$ 640	1	0.382	0.451	0.710	0.279	0.411	0.806	0.196
DuoSpaceNet [23]	1600 $\times$ 640	1	0.399	0.462	0.683	0.279	0.376	0.829	0.205
BEVFormer [30]	1600 $\times$ 900	4	0.416	0.517	0.673	0.274	0.372	0.394	0.198
PETRv2 [40]	1600 $\times$ 640	6	0.421	0.524	0.681	0.267	0.357	0.377	0.186
Sparse4D [34]	1600 $\times$ 640	4	0.436	0.541	0.633	0.279	0.363	0.317	0.177
DuoSpaceNet [23]	1600 $\times$ 640	4	0.443	0.547	0.603	0.275	0.360	0.314	0.195

Table 1: Comparison of 3D detection results on the nuScenes val set. All methods are trained for 24 epochs using ResNet-101-DCN and benefit from perspective view pre-training. Test time augmentation is not used for all experiments [23].

All the papers that have been discussed here make significant contributions to the 3D object detection task. They are targeted by this study because it takes deep scene understanding to close the loop towards planning. Sparse queries, perspective projection, temporal fusion, and real-time performance are some key elements that have proven great potential for future work, in an attempt to overcome the limitations of performance, expensive computations, and challenging scenarios.

2.2 Map Generation

Scene understanding is a requirement for the capability of motion planning in autonomous driving; hence, more than the objects in the scene, an AD* system needs contributions also from the static world that lies around the ego-vehicle. In that regard, researchers proposed various static world representation methods, which will be noted in this section.

*Autonomous Driving

2.2.1 Occupancy Grids

Mapping in its initial form was designed for robots and started from the very well-known approach of occupancy grids [45]. This is a family of algorithms that discretize the environment around an agent in cells of a predefined shape and dimension, and compute or predict for each cell a state, which can be free, occupied, or unknown. So, the map was in fact a grid-like structure composed of cells seen from a top-view perspective (or bird’s eye view). In practice, for vehicles, it has been proven that such a simple representation of cell state is not sufficient, as it loses a lot of information from the actual area represented by a cell. As a consequence, semantic occupancy grids or semantic maps were introduced, where the cells were not only occupied or free, but contained the semantic attribute, i.e., a class label representing the type of object that lies in that cell.

The task that results from this problem is the mapping of the images (usually a surround-view setup) into maps. Here, Saha et al. [43] propose a method to perform this ”translation”. It designs a mapping between image columns and BEV rays, which is learned via two types of attention: inter-plane (as in Figure 7) and per-ray self-attention. This takes into consideration that an image column can influence multiple BEV cells because the image is first sent to BEV, and then the cells are refined via attention. They also use the concept of *monotonic attention*, which is based on the fact that in an image, as it is traversed from bottom to top, the radial distance to the 3D point corresponding to an image pixel is typically increasing. Like this, attention is constrained to prevent overfitting, and it is used for MAIL (monotonic attention with infinite look-back), to understand differently the context above a point and the context below a point.

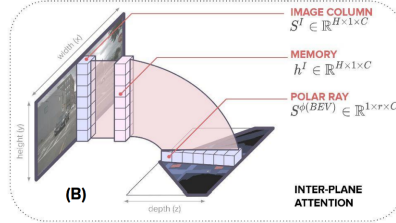


Figure 7: Inter-plane attention to translate images into maps [43].

A remarkable paper in this map segmentation area is BevFormer [30], which provides a spatio-temporal transformer architecture that predicts a dense grid structure with semantic information from a set of surround view images. It involves the use of attention in a spatial manner to link the 2D image features to the 3D grid cells, and it also adds temporal attention for fusing the grid from the past frame with the current one to improve stability. For the spatial cross attention mechanism, deformable attention is used, but adapted to a 3D representation of the queries, like pillars. Then, the 2D image features are ”lifted”, as in the

previous work from Nvidia researchers [42], to a 3D space from which the 3D query, depending on the image view, as seen in figure 8, on the left side.

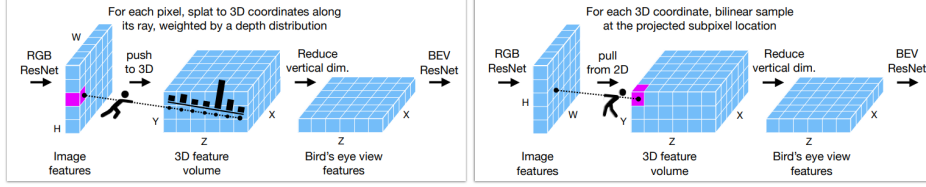


Figure 8: Comparison of 2D to 3D projection between LSS [42] and SimpleBEV [17].

To enhance the runtime of this type of image to BEV network, Adam et al. [17] propose an efficient transformation paradigm with no parameters, but a set of look-up tables. The work involves changing the lifting phase from *Lift, splat, shoot* [42] such that they construct a 3D volume of coordinates from the BEV space and project all of them back into the images to collect all 2D features that contribute to a specific BEV cell. This approach allows for avoiding introducing a depth network to contribute to a 2D to 3D projection, and saves time and model parameters. The authors also insist on the fact that cameras and radar make a good pair in a vehicle designed for autonomous driving and provide an extensive evaluation of their method’s performance with this kind of sensor setup.

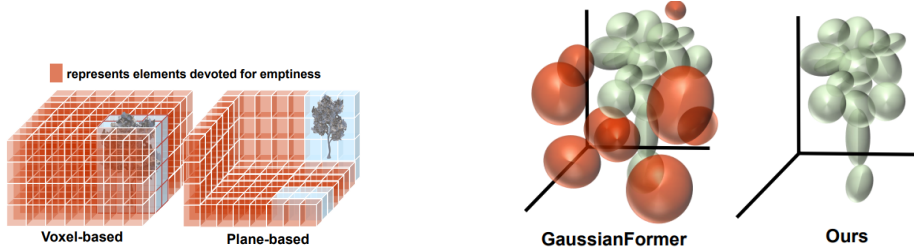


Figure 9: Representation comparison of Gaussians [21].

Another possible representation for occupancy is the *voxelized* one, which is performed in GaussianFormer [22]. This paper proposes a method for occupancy prediction that uses sparse convolutions and cross-attention to efficiently transform 2D images into 3D Gaussians. The method is rather similar to the concept of DETR-like methods [6], but with a set of queries initialized with random Gaussians that represent the spread of a 2D object in 3D, with class, approximate dimensions, and coverage in 3D. The voxelization is obtained via a Gaussian splatting method that merges all the Gaussians that contribute to a specific voxel. As an enhancement, GaussianFormer-2 [21] proposes a probabilistic Gaussian

superposition. This makes the model more efficient by avoiding unnecessary representation of overlapping Gaussians. Each Gaussian is interpreted as a probability distribution of the neighbours being occupied. Like this, no empty voxels will be represented, so no unnecessary memory is occupied. In general, using voxels is an expensive operation for autonomous driving, but these papers bring important contributions towards the real-time performance of this setup.

2.2.2 Vectorized Scene Representation

To simplify the expensive occupancy representation, literature introduced the concept of vectorized scene representation [15], where all the elements of the map are represented as vectors. It has been proven that such an approach offers a sufficiently comprehensive scene understanding, which can then be used for trajectory planning in an autonomous driving system.

HMapNet [27] introduced the use of transformers in generating online vectorized maps. The network produces three outputs: the semantic map, the instance embedding, and the direction prediction. To these, a post-processing module is added to perform clustering and non-maxima suppression for obtaining the vectorized map, which includes lane dividers, road boundaries and pedestrian crossings. These additional steps make the method lose both accuracy and convenience, which is why VectorMapNet [28] enabled the neural network seen in figure 10 to predict the vectors for the map and dynamic agents directly. This method utilizes the polyline as a uniform, geometry-consistent representation for all map elements. The network contains a feature extractor that converts from image space to BEV space, a sparse keypoint detector, and a polyline generator to produce ordered sequences of keypoints representing the map. The polylines are obtained sequentially, in a conditional manner, and the vertices have embeddings that enable the network to learn and then decode their exact position. Still, this method lacks any temporal stability mechanism, which is mandatory for a subsequent downstream task (motion prediction and planning).

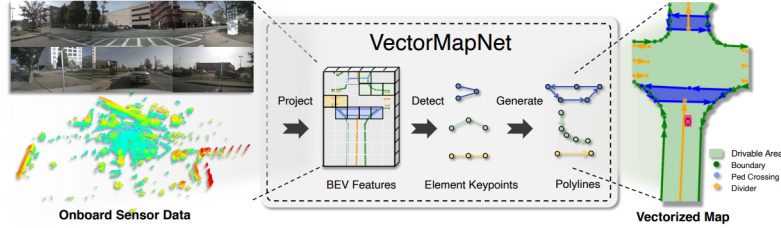


Figure 10: An overview of VectorMapNet [38].

Also specialized in online vectorized map generation, Liao et al. introduced MapTR [31] and MapTRv2 [33], efficient and fast methods that open the road towards behaviour planning in autonomous driving. They design a DETR-like

querying framework [6] accompanied by constrained equivalent permutations. The first work proved to be the first state-of-the-art online mapping, achieving real-time performance. It highlights the novelty of polyline and polygon representation of map elements as sets of points with groups of equivalent permutations to avoid ambiguity. This convention can be noted in figure 11. Map construction is formulated as a parallel regression problem, while instance matching is done via bipartite matching, and point matching is done via point2point supervision.

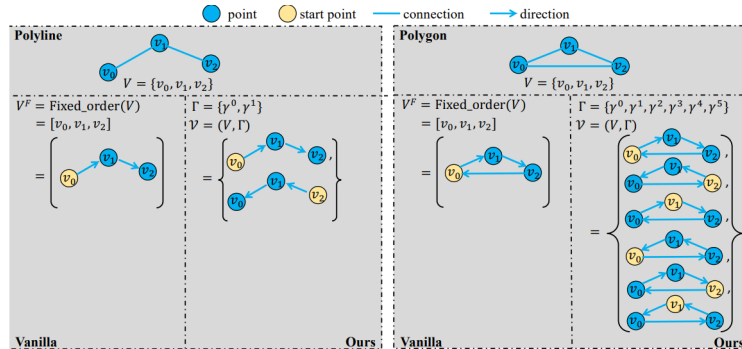


Figure 11: Illustration of permutation-equivalent shape modeling [33].

MapTRv2 comes with speed and memory enhancements by limiting the vanilla self-attention to an instance-level attention on two levels: inter-instance and intra-instance. On top of that, it imposes a physical direction for all the map elements, a feature that is crucial in motion prediction.

To further enhance the rotation and translation invariance of the map generation, Zhang et al. introduce GeMap [52] as a geometry-aware map element representation. They leverage some basic features like parallelism and perpendicularism, which are noted to be present everywhere in a driving scene (parallel lanes, obstacles perpendicular to the road surface, etc), to impose a displacement vector-based Euclidean geometry representation that penalizes the prediction via a specialized loss. Both predicted queries and ground truth data are transformed into this representation, which describes the local geometry of a map element instance using magnitudes of displacement vectors. Then, for instance, relations, angles between vectors, and magnitudes of vectors become key clues for correlation.

In table 2 adopted from [52], there is a metric comparison of the discussed online vectorized map generation methods, all transformer-based architectures that generally use camera images as sensorial input, sometimes accompanied by LiDAR point clouds for better depth estimation. The metrics that are presented in the table are based on Average Precision (AP), a standard measure in computer vision that balances prediction precision and recall.

Methods	Backbone	Dns. Loss	AP_{div} ($\uparrow$)	AP_{ped} ($\uparrow$)	AP_{bnd} ($\uparrow$)	mAP ($\uparrow$)	FPS ($\uparrow$)
VectorMapNet [38]	R50		42.5	51.4	44.1	46.0	5.3
MapTR [31]	R50		59.8	56.2	60.1	58.7	19.8
GeMap [52]	R50		65.1	59.8	63.2	62.7	19.0
HDMaNet [27]	EB0	$\checkmark$	14.4	21.7	33.0	23.0	0.7
MapTRv2 [33]	R50	$\checkmark$	68.3	68.1	69.7	68.7	15.0
GeMap [52]	R50	$\checkmark$	69.8	67.1	71.4	69.4	15.8
HDMaNet [27]	EB0 & PP		29.6	16.3	46.7	31.0	0.7
VectorMapNet [38]	R50 & PP		60.1	48.2	53.0	53.7	-
MapTR [31]	R50 & Sec		62.3	55.9	69.3	62.5	6.7
GeMap [52]	R50 & Sec		66.3	62.2	71.1	66.5	7.3
MapTRv2 [33]	R50 & Sec	$\checkmark$	65.6	66.5	74.8	69.0	6.6
GeMap [52]	R50 & Sec	$\checkmark$	69.8	68.0	73.4	70.4	6.8
MapTRv2 [33]	V2-99		73.7	71.4	75.0	73.4	10.1
GeMap [52]	Swin-T		72.8	70.4	72.8	72.0	11.4
GeMap [52]	V2-99		76.0	74.3	77.7	76.0	10.4

Table 2: Comparison on the nuScenes dataset for online vectorized map generation. Methods with two backbones utilize both camera and LiDAR inputs. “Dns. Loss” denotes whether any dense prediction (e.g., semantic segmentation) loss is adopted. The best result is highlighted in bold. All methods are reproduced on a single RTX3090 GPU to test FPS for fair comparison. Table adapted from [52].

AP_{div}	Average Precision for lane dividers (e.g., solid lines, dashed lines) that separate adjacent lanes.
AP_{ped}	Average Precision for pedestrian crossings (i.e., crosswalks).
AP_{bnd}	Average Precision for road boundaries (e.g., curbs, road edges, or non-drivable area boundaries).
mAP	Overall accuracy metric, calculated as the average of the AP scores across all evaluated map element categories.

The table also contains the reference to the backbone for the feature extractor from the images and, in the cases where two backbones are noted, from LiDAR point clouds. The end rows mention backbones that use attention for enhanced feature extraction.

Additionally, the final column in the table 2 is Frames Per Second (FPS), which is a measure of the model’s inference speed, indicating the number of maps it can construct per second. A higher FPS is essential for real-time operation in autonomous driving, hence it is crucial to adopt a computationally efficient approach. Multimodal sensor fusion leveraging a Transformer backbone significantly enhances perception robustness and accuracy by integrating complementary data cues. However, this advanced feature extraction introduces high computational demands, resulting in increased inference latency. MapTR [33] and GeMap [52] prove to be leaders with respect to this metric.

Despite significant efforts towards efficient online map construction, current methods, particularly those leveraging vectorized representations, primarily suffer from two key limitations: a lack of robust temporal aggregation and poor semantic generalization. While the simplicity and effectiveness of vectorized maps for end-to-end autonomous driving warrant further discussion in Section 4, the models fail to accurately capture or generalize to unusual traffic elements and driving hazards. Consequently, their performance and reliability remain highly susceptible to complex, unmodeled driving scenarios.

3 Motion Forecasting and Trajectory Prediction

Modern deep learning for end-to-end driving was introduced in 2016 by NVIDIA [3]. The researchers proposed a deep convolutional neural network that processed camera images and directly generated a steering angle for the wheel. The network was relatively simple; it only required images and the steering command from the vehicle’s CAN[†], and it was able to produce a corresponding future command. This paper was a new beginning for the self-driving idea, but obviously, it could only reach a limited performance because of the flat world assumption, and on the huge dependency of the amount and diversity of training data.

Two years later, Waymo came up with ChaffeurNet [2], a more sophisticated recurrent neural network that inputs partially processed images, as a top-view

[†]CAN (Controller Area Network) is the vehicle’s standard serial bus protocol that allows microcontrollers and Electronic Control Units (ECUs) to communicate without a host computer.

representation that emphasizes vehicles, traffic light states, and road information, and predicts waypoints that form a trajectory for the ego-vehicle, as in figure 12. This paper already introduced the idea of combined perception and planning training.

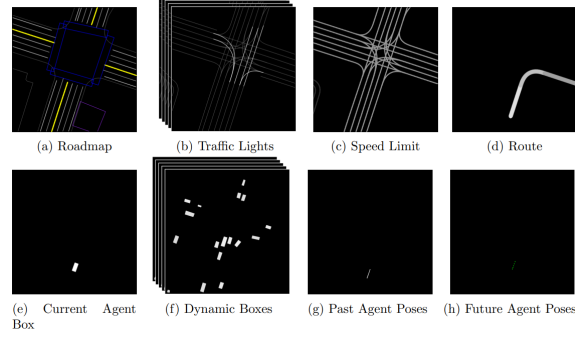


Figure 12: Driving model inputs (a-g) and output (h) [2].

Imitation learning was empowered via CARLA [14], a simulation environment that gives access to different driving scenarios and offers freedom in configuring it. Hence, a conditional approach [12] proposed the idea of using the driving command ("turn left", "turn right", "go straight") as input to the CNN for offering extra guidance for the agent. Similarly, Chen et al. [7, 8] elaborated on the idea of using imitation learning to predict waypoints from image inputs, with a top-view representation of the environment as in figure 12 and a teacher-student network that learns the output in stages. On top of these, in the same simulated environment, TransFuser [11] comes with an elevated method of imitation learning that includes sensor fusion (camera and LiDAR) and a transformer module that enables better modeling of different scene entities for enhanced driving performance. The architecture is displayed in figure 13, and it imposes feature transfer between the two input streams at different pyramid levels in the extraction phase.

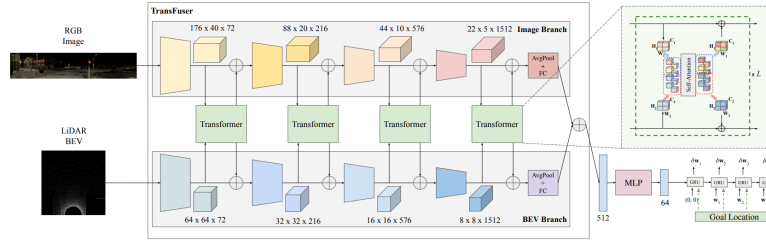


Figure 13: TransFuser overall architecture [11].

Another transformer encoder-decoder pair is proposed by WayFormer [41],

which encodes the whole driving scene in tokens and leverages the attention capabilities to model interaction between them. The authors explore mechanisms to enhance attention runtime, i.e., multi-axis attention (both in space and time), factorized attention, and latent query attention. The trajectory is predicted as a hierarchical Gaussian representation, and the one closest to the ground truth is chosen as the final prediction.

Exploiting the vectorized scene representation paradigm, HTPR [51] focuses on a way for efficient modelling of agent interactions. They introduce a module called KNARPE, which limits interaction to the close vicinity of each agent. Later, TPK [1] is further developed on HTPR with a kinematic feasibility module that proposes to enhance motion forecasting. The agent interaction layer aims to model all the agent classes, not only vehicles, as it used to be in most of the papers. This favours interaction based on a principle closer to human intuition, which considers the most relevant object for the ego vehicle to interact with, based on scores. For the kinematic model, the paper also uses the notion of control, which dictates how an agent will move. They imply kinematic models to constrain the possible trajectories that an agent can take. The detailed architecture is visible in figure 14.

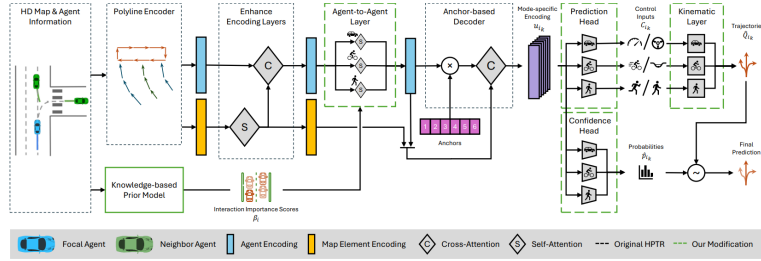


Figure 14: TPK overall architecture [1].

VADv2 [9] proposes a probabilistic planning approach. This implies generating a set of possible trajectories alongside their probability, a convention that enhances the model’s capability to better generalize and see beyond the most present trajectory from the dataset, which is usually a straight one, for in-lane driving. The action space is discretized in a planning vocabulary, and based on these and on a large set of expert demonstrations, an action distribution along a probability score is generated for comparison with the ground truth. This approach demonstrates a high score in closed-loop planning in CARLA, but it has not been validated on real-world data.

Hydra-MDP [29] builds upon VADv2 [25] and TransFuser [11] and proposes a multi-modal input (adding LiDAR) and a *knowledge distillation* setup, where the student only sees the raw sensor information, while the teacher has access to perception outputs from simulations, so that a large number of challenging scenarios are covered, enhancing the generalization capabilities for this planning

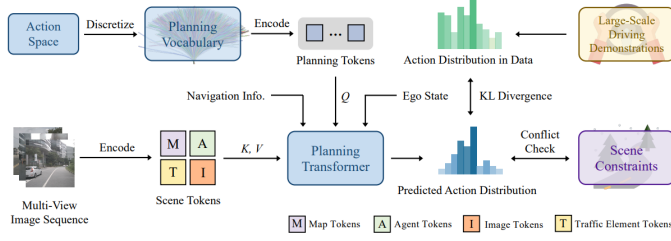


Figure 15: VADv2 overall architecture [9].

task.

Following the modern generative methods inspired by LLMs[‡], GenAD [53] formulates the problem of planning as a future generation one. The first part of the proposed model is based on BEVFormer [30], the second one on VAD [25], and the generative part makes use of self-attention on instance tokens. This means that all the instances (agents) in the map will interact with each other, with the map, and with themselves, and then also with the encoded ground truth trajectories to obtain Gaussian distributions. The generation is done waypoint by waypoint, and uses GRUs to model the temporal evolution of the map elements and traffic participants, as illustrated in figure 16. This paper mainly compares its performance with VAD, which will be discussed in the next section, and proposes, as future work, adversarial networks and diffusion models.

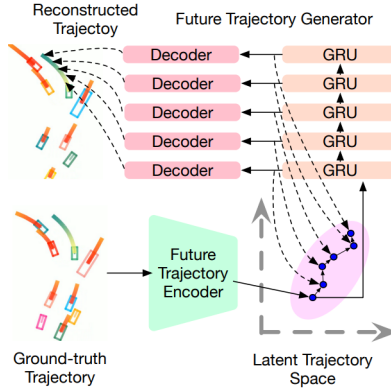


Figure 16: Illustration of the proposed trajectory prior modeling and future generation [53].

Diffusion policy [10] has also been explored as a planning technique as a two-step process of learning a final result. Noising and denoising are two opposite

[‡]Large Language Models

processes that imply construction and reconstruction of a structure, in this case, a trajectory of the movements that a robot must perform or the waypoints in the path that a vehicle must follow. Still, the focus is on the denoising phase, which is proposed in this paper, an iterative process that, following a policy based on sensorial input (camera images), succeeds in generating a reliable sequence of refined actions for the robot. Denoising is the exact reverse of noise; hence, the proposed network does not predict directly refined actions, but noise that is then subtracted from the noisy measurements. The authors propose two neural networks, one with CNNs and one with a transformer module, which uses the sequential attribute of the action space that the robot must perform. An immediate user of diffusion is MotionDiffuser [26], a model capable of including perception data (ego poses, objects, traffic elements) in the task of motion prediction for multiple traffic users. Besides the denoiser, the model also includes a transformer that tokenizes traffic data and uses it as conditional input both in training and at inference. At training, the denoiser learns to reconstruct the ground truth trajectory to which random noise has been added. At inference, random trajectories are generated for all the agents, then the denoiser, together with the observations, produces the final refined output. The overview can be seen in figure 17.

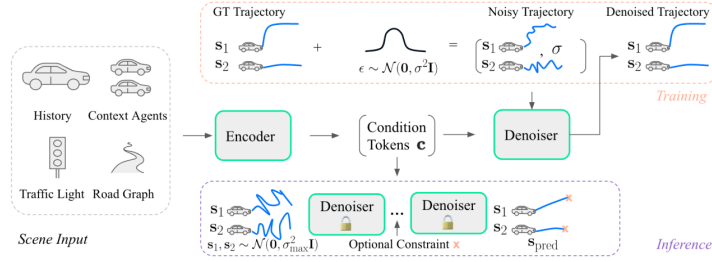


Figure 17: An overview of MotionDiffuser architecture [26].

4 Planning-Oriented Autonomous Driving

This paradigm of planning-oriented autonomous driving implies having a semi-modular trajectory prediction pipeline that uses intermediary representations of the environment (queries and feature maps) to generate a set of waypoints for the ego vehicle.

Such a model was presented by Hu et al. as ST-P3 [18], designed to simultaneously train perception, prediction, and planning in a spatio-temporal setup. They provide a mechanism to accumulate image features via 3D lifting [42] to build BEV maps with multiple traffic elements, and a temporal fusion method that uses gated recurrent units (GRUs) to accumulate perception data along a period of time, on 2 pathways: one that counts the past, and one that counts the uncertainties in the

scene. The future trajectory generation utilizes an aggregated cost (learning-based and policy-based) that samples probabilistic paths, from which one is selected.

Another alternative for a similar problem statement, in [24], the driving commands are directly generated via a knowledge distillation setup, where the teacher supervision comes as expert BEV features for the intermediary scene representation. The authors also implement a multi-modal input provisioning, as a camera-LiDAR fusion for enhanced depth perception.

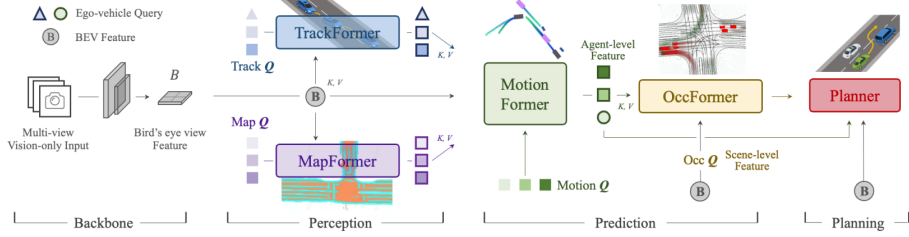


Figure 18: An overview of the query-based UniAD architecture [20].

From the same conference, CVPR 2023, a remarkable paper that proposes a unified and comprehensive planning-oriented paradigm setup is UniAD [20]. The network the authors propose can be observed in figure 18. It is complex and transformer-based, containing relative ADAS modules that communicate with each other at feature space level, via queries. The most interesting part of the proposed approach is the prediction stage, where motion queries, which model interaction between agents and map elements, are fused in an occupancy map, modeled as a dense grid with binary values that represent the state of each cell for a set of consecutive future frames. Occupancy is further used in combination with the ego-vehicle motion query to model the planned trajectory, assisted by a high-level driving command that is turned into an embedding and ingested in the planning module. This paper opens an important gate towards planning-oriented autonomous driving, but in its initial state, it is not suitable for a real-time system deployment and suffers from distance estimation accuracy.

To improve computational efficiency, VAD [25] avoids dense occupancy grids by proposing a vectorized scene representation of the driving environment. As illustrated in Figure 19, the architecture is a query-based framework that identifies abstract map elements and dynamic agent motion tracks. By modeling complex interactions among these vectorized elements and the ego vehicle, VAD generates a constrained trajectory, surpassing UniAD in performance and setting a new benchmark for real-time end-to-end autonomous driving, at that moment in time. The measurements are available in table 3.

SparseAD [50] follows the sparse detection methods as PETR [39] and uses a fully sparse query paradigm, where a multi-task architecture is proposed to elaborate each of the previously mentioned AD tasks, and to emphasize the way

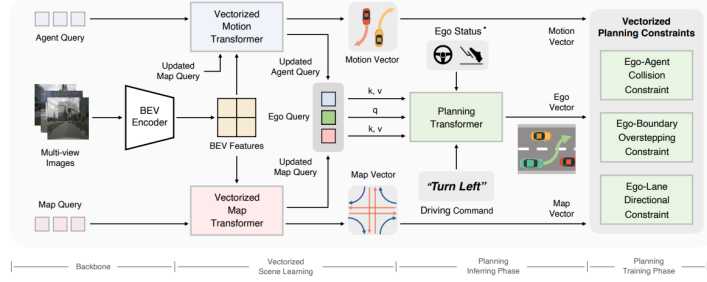
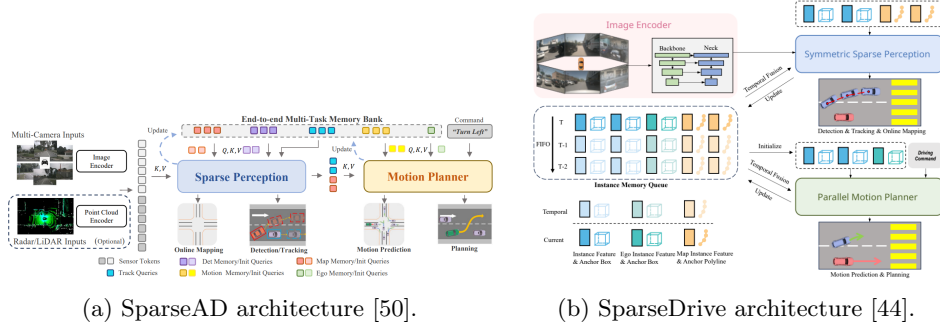


Figure 19: An overview of the VAD architecture [25].

these intermediate representations of different traffic-relevant objects interact to fulfil the end-to-end goal, while introducing radar and LiDAR information to enhance 3D perception. They enforce temporal supervision via a so-called end-to-end memory bank, as seen in figure 20a, and propose an online mapping algorithm that does not need a BEV grid representation, so it completely avoids any dense processing. Similarly, SparseDrive [44] unifies detection, tracking, and online mapping in a symmetric structure that models the driving scene. It proposes a parallel motion prediction and planning submodule in which the ego agent is initialized with the smallest feature map from the front camera, which offers a small dimension and a high amount of encoded information. The spatio-temporal interactions between agents and the map occur via a memory queue, also seen in figure 20b, and the trajectory generation is similar to what VAD [25] does with hierarchical planning. SparseDrive is proven to be the fastest end-to-end driving approach from this study, as stated in table 3.



ParaDrive [48] also attempts to address real-time efficiency and proposes a parallel architecture that operates on a BEV grid using cross-attention, as seen in figure 21. The authors highlight the power of designing independent modules on top of UniAD [20] prior work via a cross-attention mechanism that keeps the consistency between the different tasks, without the explicit need of waiting for

the previous task to generate the intermediate results. Still, as it also produces occupancy, the FPS measure is not the best, as from table 3.

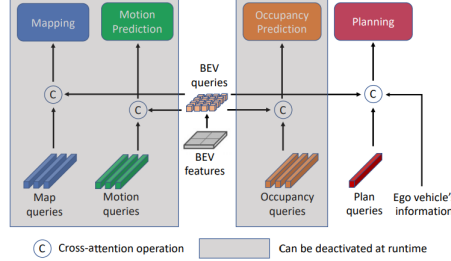


Figure 21: An overview of the ParaDrive architecture [48].

From the same authors of GaussianFormer [22], GaussianAD [49] proposes a comprehensive framework for end-to-end autonomous driving, which models the world using 3D Gaussians as a sparse representation that contains irregular obstacles. It offers a wider view of the scene compared to the fully sparse methods, which only refer to bounding boxes and map elements, and do not rely on voxels to avoid the high computational effort needed. On top of the original paper, this work adds temporal information and applies a motion prediction module via cross-attention to determine future trajectories for all the objects, followed by a planning module to generate the final trajectory for the ego agent. The authors also propose a flow module that predicts the future occupancy from the current one. The main limitation of this paper is that GaussianAD cannot predict accurate scene evolution due to not considering the newly observed areas.

Then, in the spirit of the early discussed diffusion policy [10], authors from TransFuser [11] propose DiffusionDrive [32], a combination of their previous work and VADv2 [9], which brings run-time improvements and better performance in complex situations. One of the main contributions of the paper is the diffusion decoder, a transformer module inside the model that is capable of refining the initial trajectories predicted as anchored Gaussian distributions to which noise has been added. The module uses cross attention, first within the generated map view (BEV) and then to interact with the localization data of the agent. Then, a feed-forward network is integrated, followed by a Timestep Modulation layer and an MLP. The aim is to predict a denoised trajectory for each one of the noisy ones, based on a policy that includes the classical driving patterns. This denoising process is cascaded multiple times, depending on the available resources, and, in the end, the trajectory with the highest score is selected as output. With this paper, end-to-end autonomous driving reaches successful performance and real-time running on a GPU, noted in table 3.

Table 3: Open-Loop Planning Performance on the nuScenes Validation Set (Camera-Only)

Method	Conf./Year	L2 Error (m) ↓				Collision Rate (%) ↓				FPS ↑
		1s	2s	3s	Avg.	1s	2s	3s	Avg.	
ST-P3	ECCV '22	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71	1.6
UniAD	CVPR '23	0.45	0.70	1.04	0.73	0.62	0.58	0.63	0.61	1.8
VAD	ICCV '23	0.41	0.70	1.05	0.72	0.03	0.19	0.43	0.21	4.5
SparseDrive-B	ICRA '24	0.29	0.55	0.91	0.58	0.01	0.02	0.13	0.06	7.3
ParaDrive	CVPR '24	0.50	1.38	2.76	1.55	0.08	0.15	0.30	0.18	5.0
GenAD	ArXiv '24	0.36	0.83	1.55	0.91	0.06	0.23	1.00	0.43	4.6
GaussianAD	ArXiv '24	0.40	0.64	0.88	0.64	0.09	0.38	0.81	0.42	2.5
SparseAD-B	ArXiv '24	0.15	0.31	0.57	0.35	0.00	0.06	0.21	0.09	3.5
DiffusionDrive	CVPR '25	0.27	0.54	0.90	0.57	0.03	0.05	0.16	0.08	8.2

5 Future Research Directions

In the context of planning-oriented autonomous driving, significant work has been done. Transformer-based architectures have shown impressive performance in the planning task, as they are able to incorporate information from multiple levels of abstraction and take advantage of a query-like representation of the intermediate features, which can be propagated along the autonomous driving tasks in an end-to-end system.

In this work, we explored multiple transformer-based networks and different types of attention adaptations for runtime improvements. We kept the view towards the planning goal, and we have identified several possible directions based on the noted papers in this study.

5.1 Sensor Fusion

The video camera is the foundational sensor in computer vision systems for autonomous driving, primarily due to its cost-effectiveness, versatility, and rich semantic information analogous to human visual perception. However, systems relying on monocular vision inherently struggle with accurate 3D geometric estimation of the real-world scene, and while employing multi-camera setups solves this, it demands significantly increased computational resources for feature processing and fusion. Traditionally, LiDAR sensors have been successfully utilized in combination with cameras to provide robust depth information; nevertheless, their prohibitive cost restricts their prevalence primarily to development and testing platforms. Consequently, the growing inclusion of radar units in commercial vehicle configurations positions radar as a naturally suitable sensor for achieving reliable perception.

Radars have already been used in semantic maps by LSS [42], in perspective

projection strategies by SimpleBEV [17], and even in end-to-end autonomous driving, by SparseAD [50], proving important 3D capabilities that are worth further research.

Future work will be directed towards integrating radar data with the existing camera-based vectorized scene representation. The primary challenge lies in effectively leveraging radar’s strengths, low-latency and distance measurements to compensate for the camera’s weaknesses (occlusion, fog, low light). We will target enhancing scene consistency and performance in adverse traffic conditions, while attempting to maintain a decent system runtime.

5.2 Network Optimization for Real-Time Capabilities

While Table 3 offers a quantitative comparison of Frames Per Second (FPS) across several end-to-end autonomous driving architectures, these metrics remain substantially below the threshold required for real-time series vehicle deployment. This performance ceiling is exacerbated by the constraint that production vehicles cannot rely on the high processing capacity of powerful GPUs. Consequently, future work aims to explore runtime optimization strategies for these approaches, synthesizing the most effective techniques discussed in this study, including vectorized scene representation, volume-based perspective projection, improved attention mechanisms, and parallel architectures. This optimization objective of the thesis will be a continuous task during all the experiments performed for the end-to-end autonomous driving architecture.

5.3 Challenging Scenarios

While most end-to-end evaluations cited in this study leverage well-established benchmarks like the nuScenes dataset [5] and controlled simulation environments, there remains a significant gap in validating performance across the most challenging, high-dynamic driving scenarios. Given that this research targets highly dynamic situations, future work will prioritize efficiently modeling the numerous dynamic traffic agents and the unpredictable appearance of unstructured obstacles widely present in urban driving. To design a more robust trajectory planning pipeline in these challenging environments, we will explore methods for uncertainty and risk assessment. Specifically, this includes investigating diffusion and Gaussian modeling approaches to accurately quantify and propagate prediction uncertainty, leading to demonstrably safer decision-making under high-risk, unstructured conditions.

6 Final Remarks

Given the extensive work that has already been released in the literature, there is a substantial pool of techniques that may be applied for future work associated with

this thesis. It is widely acknowledged that neither perception nor planning represents a fully solved problem in the autonomous driving domain. Consequently, significant scope remains for innovative architectural synthesis and experimental validation to demonstrably improve performance, particularly by addressing the critical trade-offs between accuracy, safety, and real-time efficiency.

References

- [1] Marius Baden, Ahmed Abouelazm, Christian Hubschneider, Yin Wu, Daniel Slieter, J. Marius Zöllner, Tpk: Trustworthy trajectory prediction integrating prior knowledge for interpretability and kinematic feasibility, 2025, available from: <https://arxiv.org/abs/2505.06743>.
- [2] Mayank Bansal, Alex Krizhevsky, Abhijit Ogale, Chauffeurnet: Learning to drive by imitating the best and synthesizing the worst, *arXiv*, 2018.
- [3] Mariusz Bojarski, Davide Del Testa, Daniel Dworakowski, Bernhard Firner, Beat Flepp, Praseoon Goyal, Lawrence D. Jackel, Mathew Monfort, Urs Muller, Jiakai Zhang, Xin Zhang, Jake Zhao, Karol Zieba, End to end learning for self-driving cars., *CoRR*, 2016.
- [4] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, Oscar Beijbom, nuscenes: A multimodal dataset for autonomous driving, 2020, available from: <https://arxiv.org/abs/1903.11027>.
- [5] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, Oscar Beijbom, nuscenes: A multimodal dataset for autonomous driving, in *CVPR*, 2020.
- [6] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, Sergey Zagoruyko, End-to-end object detection with transformers, 2020, available from: <https://arxiv.org/abs/2005.12872>.
- [7] Dian Chen, Philipp Krähenbühl, Learning from all vehicles, in *CVPR*, 2022.
- [8] Dian Chen, Brady Zhou, Vladlen Koltun, Philipp Krähenbühl, Learning by cheating, in *Conference on Robot Learning (CoRL)*, 2019.
- [9] Shaoyu Chen, Bo Jiang, Hao Gao, Bencheng Liao, Qing Xu, Qian Zhang, Chang Huang, Wenyu Liu, Xinggang Wang, Vadv2: End-to-end vectorized autonomous driving via probabilistic planning, *arXiv preprint arXiv:2402.13243*, 2024.

- [10] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin CM Burchfiel, Shuran Song, Diffusion Policy: Visuo-motor Policy Learning via Action Diffusion, in *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023, <https://doi.org/10.15607/RSS.2023.XIX.026>.
- [11] Kashyap Chitta, Aditya Prakash, Bernhard Jaeger, Zehao Yu, Katrin Renz, Andreas Geiger, Transfuser: Imitation with transformer-based sensor fusion for autonomous driving, *Pattern Analysis and Machine Intelligence (PAMI)*, 2023.
- [12] Felipe Codevilla, Matthias Müller, Antonio López, Vladlen Koltun, Alexey Dosovitskiy, End-to-end driving via conditional imitation learning, year, in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018, <https://doi.org/10.1109/ICRA.2018.8460487>.
- [13] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, 2021, available from: <https://arxiv.org/abs/2010.11929>.
- [14] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, Vladlen Koltun, CARLA: An open urban driving simulator, in Sergey Levine, Vincent Vanhoucke, Ken Goldberg (Eds.), *Proceedings of the 1st Annual Conference on Robot Learning*, Volume 78 of *Proceedings of Machine Learning Research*. year, PMLR, 13–15 Nov 2017, available from: <https://proceedings.mlr.press/v78/dosovitskiy17a.html>.
- [15] Jiyang Gao, Chen Sun, Hang Zhao, Yi Shen, Dragomir Anguelov, Congcong Li, Cordelia Schmid, Vectornet: Encoding hd maps and agent dynamics from vectorized representation, year, in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, <https://doi.org/10.1109/CVPR42600.2020.01154>.
- [16] Autonomous Vision Group, Self-driving cars — lecture, <https://uni-tuebingen.de/en/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/self-driving-cars/>, 2024.
- [17] Adam W. Harley, Zhaoyuan Fang, Jie Li, Rares Ambrus, Katerina Fragkiadaki, Simple-BEV: What really matters for multi-sensor bev perception?, in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

- [18] Shengchao Hu, Li Chen, Penghao Wu, Hongyang Li, Junchi Yan, Dacheng Tao, St-p3: End-to-end vision-based autonomous driving via spatial-temporal feature learning, in *European Conference on Computer Vision (ECCV)*, 2022.
- [19] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqu Chai, Senyao Du, Tianwei Lin, Wenhai Wang, Lewei Lu, Xiaosong Jia, Qiang Liu, Jifeng Dai, Yu Qiao, Hongyang Li, Planning-oriented autonomous driving, year, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2023.
- [20] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqu Chai, Senyao Du, Tianwei Lin, Wenhai Wang, Lewei Lu, Xiaosong Jia, Qiang Liu, Jifeng Dai, Yu Qiao, Hongyang Li, Planning-oriented autonomous driving, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [21] Yuanhui Huang, Amonnut Thammatadatrakoon, Wenzhao Zheng, Yunpeng Zhang, Dalong Du, Jiwen Lu, Gaussianformer-2: Probabilistic gaussian superposition for efficient 3d occupancy prediction, year, in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [22] Yuanhui Huang, Wenzhao Zheng, Yunpeng Zhang, Jie Zhou, Jiwen Lu, Gaussianformer: Scene as gaussians for vision-based 3d semantic occupancy prediction, in *Computer Vision – ECCV 2024: 18th European Conference, Milan, Italy, September 29–October 4, 2024, Proceedings, Part XXVII*, Springer-Verlag, Berlin, Heidelberg, 2024, available from: https://doi.org/10.1007/978-3-031-73383-3_22.
- [23] Zhe Huang, Yizhe Zhao, Hao Xiao, Chenyan Wu, Lingting Ge, Duospacenet: Leveraging both bird’s-eye-view and perspective view representations for 3d object detection, year, in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025.
- [24] Xiaosong Jia, Penghao Wu, Li Chen, Jiangwei Xie, Conghui He, Junchi Yan, Hongyang Li, Think twice before driving: Towards scalable decoders for end-to-end autonomous driving, in *CVPR*, 2023.
- [25] Bo Jiang, Shaoyu Chen, Qing Xu, Bencheng Liao, Jiajie Chen, Helong Zhou, Qian Zhang, Wenyu Liu, Chang Huang, Xinggang Wang, Vad: Vectorized scene representation for efficient autonomous driving, year, in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [26] Chiyu Max Jiang, Andre Cornman, Cheolho Park, Ben Sapp, Yin Zhou, Dragomir Anguelov, Motiondiffuser: Controllable multi-agent motion prediction using diffusion, 2023.

- [27] Qi Li, Yue Wang, Yilun Wang, Hang Zhao, Hdmapnet: An online hd map construction and evaluation framework, year, in *2022 International Conference on Robotics and Automation (ICRA)*, 2022, <https://doi.org/10.1109/ICRA46639.2022.9812383>.
- [28] Qi Li, Yue Wang, Yilun Wang, Hang Zhao, Hdmapnet: An online hd map construction and evaluation framework, in *2022 International Conference on Robotics and Automation (ICRA)*. year, IEEE Press, 2022, available from: <https://doi.org/10.1109/ICRA46639.2022.9812383>.
- [29] Zhenxin Li, Kailin Li, Shihao Wang, Shiyi Lan, Zhiding Yu, Yishen Ji, Zhiqi Li, Ziyue Zhu, Jan Kautz, Zuxuan Wu, Yu-Gang Jiang, Jose M. Alvarez, Hydra-mdp: End-to-end multimodal planning with multi-target hydra-distillation, 2024, available from: <https://arxiv.org/abs/2406.06978>.
- [30] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, Jifeng Dai, Bevformer: Learning bird’s-eye-view representation from multi-camera images via spatiotemporal transformers, in *Computer Vision – ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part IX*, Springer-Verlag, Berlin, Heidelberg, 2022, p. 1–18.
- [31] Bencheng Liao, Shaoyu Chen, Xinggang Wang, Tianheng Cheng, Qian Zhang, Wenyu Liu, Chang Huang, Maptr: Structured modeling and learning for online vectorized HD map construction, in *The Eleventh International Conference on Learning Representations*, 2023, available from: https://openreview.net/forum?id=k7p_YA07yE.
- [32] Bencheng Liao, Shaoyu Chen, Haoran Yin, Bo Jiang, Cheng Wang, Sixu Yan, Xinbang Zhang, Xiangyu Li, Ying Zhang, Qian Zhang, Xinggang Wang, Diffusiondrive: Truncated diffusion model for end-to-end autonomous driving, *arXiv preprint arXiv:2411.15139*, 2024.
- [33] Bencheng Liao, Shaoyu Chen, Yunchi Zhang, Bo Jiang, Qian Zhang, Wenyu Liu, Chang Huang, Xinggang Wang, Maptrv2: An end-to-end framework for online vectorized hd map construction, *International Journal of Computer Vision*, pp. 1–23, 2024.
- [34] Xuewu Lin, Tianwei Lin, Zixiang Pei, Lichao Huang, Zhizhong Su, Sparse4d: Multi-view 3d object detection with sparse spatial-temporal fusion, 2022.
- [35] Xuewu Lin, Tianwei Lin, Zixiang Pei, Lichao Huang, Zhizhong Su, Sparse4d v2: Recurrent temporal fusion with sparse model, 2023.
- [36] Xuewu Lin, Zixiang Pei, Tianwei Lin, Lichao Huang, Zhizhong Su, Sparse4d v3: Advancing end-to-end 3d detection and tracking, 2023.

- [37] Haisong Liu, Yao Teng, Tao Lu, Haiguang Wang, Limin Wang, Sparsebev: High-performance sparse 3d object detection from multi-camera videos, 2023, available from: <https://arxiv.org/abs/2308.09244>.
- [38] Yicheng Liu, Tianyuan Yuan, Yue Wang, Yilun Wang, Hang Zhao, Vectormapnet: end-to-end vectorized hd map learning, in *Proceedings of the 40th International Conference on Machine Learning, ICML'23*. JMLR.org, 2023.
- [39] Yingfei Liu, Tiancai Wang, Xiangyu Zhang, Jian Sun, Petr: Position embedding transformation for multi-view 3d object detection, 2022, available from: <https://arxiv.org/abs/2203.05625>.
- [40] Yingfei Liu, Junjie Yan, Fan Jia, Shuailin Li, Aqi Gao, Tiancai Wang, Xiangyu Zhang, Jian Sun, Petr2: A unified framework for 3d perception from multi-camera images, 2022, available from: <https://arxiv.org/abs/2206.01256>.
- [41] Nigamaa Nayakanti, Rami Al-Rfou, Aurick Zhou, Kratarth Goel, Khaled S. Refaat, Benjamin Sapp, Wayformer: Motion forecasting via simple & efficient attention networks, year, in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, <https://doi.org/10.1109/ICRA48891.2023.10160609>.
- [42] Jonah Philion, Sanja Fidler, Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3d, in *Proceedings of the European Conference on Computer Vision*, 2020.
- [43] Avishkar Saha, Oscar Mendez, Chris Russell, Richard Bowden, Translating images into maps, in *2022 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2022.
- [44] Wenchao Sun, Xuewu Lin, Yining Shi, Chuang Zhang, Haoran Wu, Sifa Zheng, Sparsedrive: End-to-end autonomous driving via sparse scene representation, *arXiv preprint arXiv:2405.19620*, 2024.
- [45] Sebastian Thrun, Wolfram Burgard, Dieter Fox, *Probabilistic Robotics (Intelligent Robotics and Autonomous Agents)*, The MIT Press, 2005.
- [46] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, Illia Polosukhin, Attention is all you need, in I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), *Advances in Neural Information Processing Systems*, Volume 30. Curran Associates, Inc., 2017, available from: https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

- [47] Yue Wang, Vitor Guizilini, Tianyuan Zhang, Yilun Wang, Hang Zhao, Justin Solomon, Detr3d: 3d object detection from multi-view images via 3d-to-2d queries, 2021, available from: <https://arxiv.org/abs/2110.06922>.
- [48] Xinshuo Weng, Boris Ivanovic, Yan Wang, Yue Wang, Marco Pavone, Parade: Parallelized architecture for real-time autonomous driving, year, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024.
- [49] Yao Zheng Sicheng Zuo Zixun Xie Longchao Yang Yong Pan Zhihui Hao Peng Jia XianPeng Lang Shanghang Zhang Wenzhao Zheng, Junjie Wu, Gaussianad: Gaussian-centric end-to-end autonomous driving, *arXiv preprint arXiv: xxx*, 2024.
- [50] Diankun Zhang, Guoan Wang, Runwen Zhu, Jianbo Zhao, Xiwu Chen, Siyu Zhang, Jiahao Gong, Qibin Zhou, Wenyuan Zhang, Ningzi Wang, Feiyang Tan, Hangning Zhou, Ziyao Xu, Haotian Yao, Chi Zhang, Xiaojun Liu, Xiaoguang Di, Bin Li, Sparsead: Sparse query-centric paradigm for efficient end-to-end autonomous driving, 2024, available from: <https://arxiv.org/abs/2404.06892>.
- [51] Zhejun Zhang, Alexander Liniger, Christos Sakaridis, Fisher Yu, Luc Van Gool, Real-time motion prediction via heterogeneous polyline transformer with relative pose encoding, in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [52] Zhixin Zhang, Yiyuan Zhang, Xiaohan Ding, Fusheng Jin, Xiangyu Yue, Online vectorized hd map construction using geometry, *arXiv preprint arXiv:2312.03341*, 2023.
- [53] Wenzhao Zheng, Ruiqi Song, Xianda Guo, Chenming Zhang, Long Chen, Genad: Generative end-to-end autonomous driving, *arXiv preprint arXiv: 2402.11502*, 2024.
- [54] Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang Wang, Jifeng Dai, Deformable detr: Deformable transformers for end-to-end object detection, 2021, available from: <https://arxiv.org/abs/2010.04159>.